

Measurement and Analysis of Thermal Noise in Resistors: Experimental Verification of Nyquist's Prediction

J. Folk^{1,2} and Camus Wang²

¹*Quantum Matter Institute, University of British Columbia, Vancouver, British Columbia, V6T1Z4, Canada*

²*Department of Physics and Astronomy, University of British Columbia, Vancouver, British Columbia, V6T1Z4, Canada*

(Dated: December 15, 2025)

This paper reports an experimental verification of Nyquist's theory of thermal noise in conductors. Using a voltage amplifier and a sound card, we measured voltage fluctuations across resistors ranging from 0 – 30 k Ω and used cross-spectral density analysis to eliminate background instrument noise. The voltage noise density we measured was linearly proportional to the resistance. We then performed a linear fit, obtaining a slope consistent with the theoretical prediction $4k_B T$ within experimental uncertainty, and a non-negligible intercept. This leads us to conclude that the observed voltage fluctuations originate from the thermal motion of charge carriers and conform to Nyquist's prediction: $S_V = 4k_B TR[1]$, and we discuss possible sources of the intercept error.

The random motion of electrons in conductors generates voltage fluctuations. This intrinsic noise fundamentally limits the sensitivity of electrical measurements—the ability to accurately capture minute fluctuations in electrical signals—and persists even in the absence of an external power source. At the same time, a quantitative understanding of this noise provides a useful reference for evaluating the performance of electronic instrumentation. In 1928, Nyquist described this voltage fluctuation as $V^2/\Delta f = 4k_B TR[1]$, where Δf is the width of the frequency range selected in the energy summation and k_B is the Boltzmann constant. This expression shows that the thermal voltage noise depends only on the conductor's temperature and resistance, independent of any external power source. However, experimentally verifying this relation is challenging because thermal noise is often masked by the instrument's background noise. To address this, we employed a cross-spectral density (CSD) technique to isolate intrinsic noise from amplifier background[2]. In this work, we designed electrical circuits and measured the voltage noise under seven different conditions (short circuit $\approx 0 \Omega$, 330 Ω , 1 k Ω , 10 k Ω , 12 k Ω , 15 k Ω , 30 k Ω) to experimentally test the validity of Nyquist's theory and to assess whether the observed voltage noise is intrinsic in origin.

In measurements, we employed a preamplifier that consists of two cascaded non-inverting amplifiers (ADA4625 and ADA4522), each incorporating a feedback capacitor for low-pass filtering. Consequently, its frequency response can be modeled as a two-pole system[3]:

$$G(f) = \frac{G_0}{\sqrt{(1 + (f/f_{p1})^2)(1 + (f/f_{p2})^2)}}, \quad (1)$$

where f_{p1} and f_{p2} are the corner frequencies. The total output voltage from the amplifier includes contributions from the input signal as well as the amplifier's input voltage noise $V_n(t)$ and input current noise $I_n(t)$. The time-domain output is expressed as:

$$V_{\text{out}}(t) = G(f)[V_{\text{in}}(t) + I_n(t)R + V_n(t)], \quad (2)$$

These three noise terms are statistically independent and uncorrelated because they originate from different physical sources. When we calculate the total power, their cross-terms average to zero, allowing the power spectral density (PSD) to be directly linearly added as [4]

$$S_{\text{out}}(f) = G^2(f) [S_I R^2 + 4k_B TR + S_V], \quad (3)$$

where S_I and S_V are the spectral densities of the amplifier's input current and voltage noise. To extract the thermal noise separately, we used the CSD method. When the resistor is connected to two independent amplifier channels simultaneously. The frequency-domain output voltages $\tilde{V}_1(f)$ and $\tilde{V}_2(f)$ are multiplied to compute the CSD[2]:

$$S_{12}(f) = \langle \tilde{V}_1(f) \tilde{V}_2^*(f) \rangle, \quad (4)$$

where $\langle \cdot \rangle$ is the expectation value. Expanding this product, the cross-correlation of all uncorrelated terms originating from different physical sources expected to be zero. Consequently, the measured CSD only reflects the PSD of the common thermal noise source:

$$S_R(f) \approx |G(f)|^2 4k_B TR. \quad (5)$$

The experimental procedure can be divided into two stages. In the first part, the amplifier's frequency response was measured in order to determine the gain function $G(f)$. A voltage divider circuit was built by using two resistors ($R_1 = 10 \text{ k}\Omega$ and $R_2 = 1 \text{ k}\Omega$) to reduce the signal from the function generator in order to avoid reaching the amplifier's power supply limit. The output and input voltages were recorded on the oscilloscope, and the measured data were fitted using Eq.(1) to obtain the $G(f)$. The experimental setup is shown in Fig. 1(a).

In the second part, the passive circuit containing the resistor was placed inside a copper shell to minimize external electromagnetic interference. The experimental configuration is shown in Fig. 1(b). The voltage signal from the closed loop was amplified and observed by the

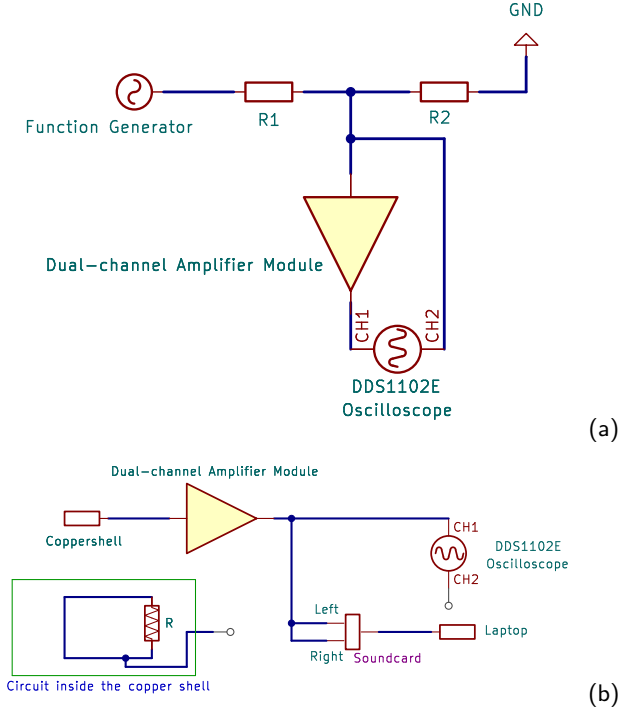


FIG. 1. Experimental schematics. (a) Voltage divider circuit used to measure the amplifier gain. An AC signal from the function generator is reduced by $R_1 = 10\text{ k}\Omega$ and $R_2 = 1\text{ k}\Omega$ before amplification. The output voltage is observed on the oscilloscope (DDS1102E), allowing $G = V_{\text{out}}/V_{\text{in}}$ to be determined as a function of frequency. (b) Set up for measuring voltage noise. The resistor under test R is enclosed in a copper shell to minimize external electromagnetic interference. The amplified outputs are observed by an oscilloscope (DDS1102E) and while recorded by a USB soundcard (ICUS-BAUDIO2d) connected to a laptop for spectral analysis. The dual-channel amplifier module internally contains two independent amplification channels, enabling cross-spectral noise measurement.

oscilloscope and the USB soundcard. Then the soundcard digitized the amplified voltage signal into WAV files at a sampling rate of 44100 Hz. However, since the soundcard does not directly measure voltage, the recorded audio data are stored as dimensionless, normalized digital values, requiring calibration to determine the actual volts represented. We recorded sinusoidal signals with varying amplitudes using the sound card. By comparing the actual peak-to-peak voltage on the oscilloscope with the peak-to-peak values in the WAV files, we confirmed that calibration-factor = $V_{pp}^{\text{osc}}/V_{pp}^{\text{wav}}$. Finally, we used Python to process the time-domain data, computing PSD and CSD, which were estimated using Welch method[5] (implemented as `signal.welch` and `signal.csd`). Then we fitted the slope of the thermal noise level $S_{12}(R)$ vs R to see if it matched the theoretical value ($4k_B T$).

By completing the first part of the experiment, we obtain $G(f)$. Completing the second part yields seven sets

of voltage-time sequence data recorded by the soundcard (short circuit $\approx 0\text{ }\Omega$, $330\text{ }\Omega$, $1\text{ k}\Omega$, $10\text{ k}\Omega$, $12\text{ k}\Omega$, $15\text{ k}\Omega$, $30\text{ k}\Omega$) and a calibration-factor of 0.0022 mV/unit . Before spectral analysis, we declare that the calibration provides a linear conversion between the sound card's digital output and the physical voltage scale and does not affect the subsequent analysis of noise scaling with resistance. First, we plot the PSD in the frequency domain and observe whether it matches the theoretical prediction without removing the amplifier's noise. Then, according to Eq.(5), we correct the data to obtain the thermal noise $S_R(f)$ and observe whether the CSD plot matches the theoretical prediction after removing the amplifier noise. Then we extract the average PSD of the thermal noise from the corrected $S_R(f)$. Since thermal noise is a randomly fluctuating noise signal, we confirm that its uncertainty mainly comes from statistical error. The uncertainty formula is $\sigma_{\langle V \rangle} = \frac{\sigma}{\sqrt{N_{\text{eff}}}}$ where σ represents the standard deviation of the raw voltage signal, and N_{eff} denotes the number of statistically independent uncorrelated samples. As a result, we plot $S(R)$ against the resistance R and fit it with the straight line model $aR+b$. By comparing the slope a obtained from the fit with the theoretical value $4k_B T$, we can qualitatively assess the consistency between the measured thermal noise and the theoretical prediction.

The measured Gain vs frequency response of the amplifier is shown in Fig. 2. The data are well described by the model of Eq(1), yielding fitted parameters $G_0 = 1496.1 \pm 150.6$, $f_{p1} = 1.05 \times 10^4 \pm 2.09 \times 10^3\text{ Hz}$, and $f_{p2} = 6.42 \times 10^4 \pm 1.38 \times 10^4\text{ Hz}$. The chi-squared $\chi^2/\text{dof} = 0.035$, which indicates that the uncertainty is overestimated at low frequencies. This overestimate is due to our conservative definition of the voltage uncertainty as half of the peak-to-peak voltage fluctuation, which increases significantly with increasing output voltage. However, the residuals show no systematic deviation across the measured frequency range, which is a good enough fit, and which confirms that Eq.(1) can capture the amplifier frequency response and be used for gain correction of subsequent studies.

To show the results and avoid repetition, this paper presents the PSD and CSD results at $R = 330\text{ }\Omega$. Fig. 3(a) shows the noise spectrum in the mid-frequency range (5000–15000 Hz) before gain correction. The PSD is calculated from the left channel signal of the sound card using the Welch method, while the CSD is the result obtained after processing the dual-channel signal using Eq.(5). Both spectral densities show significant deviations from theoretical prediction. However, it is clear that the noise spectrum amplitude obtained after CSD processing is lower than the corresponding PSD result. This indicates that the CSD method effectively suppresses uncorrelated noise components from the amplifier. Fig. 3(b) shows the noise spectrum after gain correction for the PSD and CSD results in Fig. 3(a). After

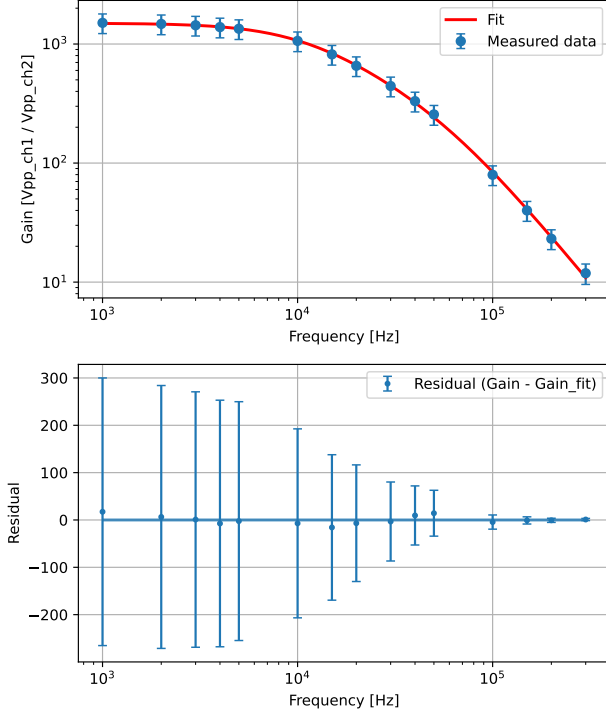


FIG. 2. Measured amplifier gain $G(f)$ (points) and best fit to Eq(1) (solid line). The fitted parameters are $G_0 = 1495.84 \pm 139.06$, $f_{p1} = 10488.18 \pm 1997.55 \text{ Hz}$, $f_{p2} = 63990.31 \pm 13571.48 \text{ Hz}$, and $\chi^2/\text{dof} = 0.037$. The lower plot shows the residuals ($G_{\text{data}} - G_{\text{fit}}$), which indicate no systematic deviation. Hence the Eq(1) accurately describes the amplifier frequency response.

gain correction, the overall amplitude of the noise spectrum corresponding to PSD is still systematically higher than the theoretical prediction. The noise spectrum corresponding to CSD exhibits approximately frequency-independent behavior across the entire mid-frequency range, and the theoretical prediction is near the midline of its noise fluctuation, indicating that further quantitative analysis using linear fitting is possible. Overall, Fig. 3 illustrates why we must use the CSD method and gain correction instead of directly using the PSD method for fitting.

To qualitatively assess the consistency between noise fluctuations and theoretical predictions, we took the average CSD of each dual channel under seven different resistances, fitted the slope using a linear model, and used the slope $a = (1.586 \pm 0.003) \times 10^{-20} \text{ V}^2/\text{Hz}/\Omega$ and $4k_B T = 1.580 \times 10^{-20} \text{ V}^2/\text{Hz}/\Omega$ with $\chi^2 = 4$. This indicates that we have underestimated the uncertainty, but based on the fitting results and values in Fig. 4, we consider this still a good enough fit.

In this experiment, we systematically analyzed thermal noise using left-channel PSD and dual-channel CSD, and corrected all results for gain using the fitted amplifier frequency response function $G(f)$. Overall, the

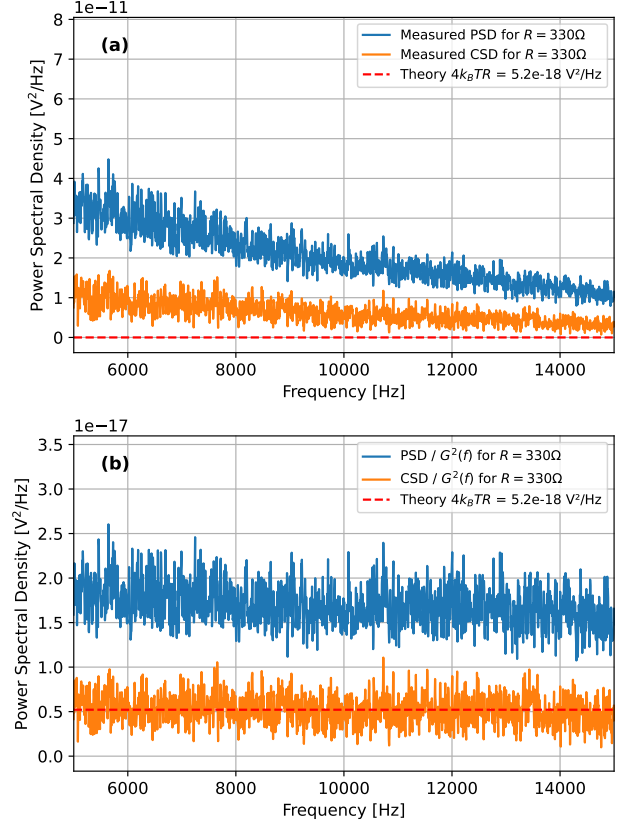


FIG. 3. Measured output power spectral density under the conditions of $T = 286.15 \text{ K}$, $k_B = 1.38 \times 10^{-23} \text{ J/K}$, $R = 330 \Omega$, and mid-frequency range $5000 - 15000 \text{ Hz}$. (a) Measured output power spectral density $S_{\text{out}}(f)$ obtained using the PSD (left channel) and CSD (dual channel) methods before gain correction. The red dashed line near 0 represents the theoretical prediction value of $4k_B T R$. (b) This shows the data processed by PSD and CSD after gain correction, $S_{\text{out}}(f)/G^2(f)$ and $S_{12}(f)/G^2(f)$. After gain correction, the CSD spectrum becomes approximately frequency-independent and agrees well with the theoretical prediction. The red dashed line near $0.5 \times 10^{-17} \text{ V}^2/\text{Hz}$ represents the theoretical prediction value of $4k_B T R$.

experimental data showed good stability after removing the uncorrelated noise by the CSD method and quantitatively remained consistent with Nyquist's prediction of thermal noise. We also argue that our results are not limited by temperature because the experiment was conducted in a room with constant temperature and humidity, and the current was so weak that it did not generate significant heat in the resistor, thus the temperature response of the system is not correlated. However, our experimental setup and results had some non-negligible limitations. Regarding the experimental setup, although a dual-channel amplifier was used and a CSD was employed to remove uncorrelated noise as much as possible, the amplifier's internal structure was not investigated in depth. Regarding the experimental results, we observed

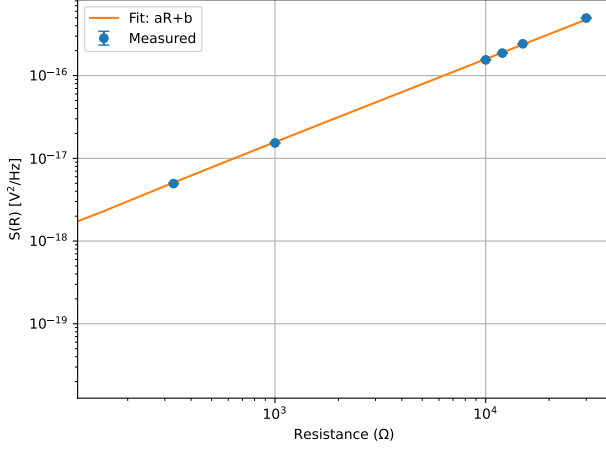


FIG. 4. This shows the measured noise strength $S(R)$ and its linear fitting results under different resistance values (log-log plot). The experimental data were fitted using the model $S(R) = aR + b$, the resulting slope was $a_{\text{exp}} = (1.586 \pm 0.003) \times 10^{-20} \text{ V}^2/\text{Hz}/\Omega$, and theoretical coefficient $4k_B T = 1.580 \times 10^{-20} \text{ V}^2/\text{Hz}/\Omega$. The corresponding chi-square value of the slope and the theoretical coefficient is $\chi^2 = 4$. The figure also shows a non-negligible intercept of $b = (-1.42 \pm 0.31) \times 10^{-19} \text{ V}^2/\text{Hz}$.

a non-negligible intercept of the linear fit for resistance. According to theoretical predictions, the voltage noise should also be zero when the resistance is zero. There are several possible reasons for this intercept, such as the resistance not being zero under short-circuit condition in the experiment, or the CSD not completely removing the intrinsic noise of the amplifier. Therefore, we believe that in future studies, we can increase the measurement of more different resistances, further verify the temperature dependence of thermal noise, or conduct an in-depth study on the internal structure of the amplifier to more accurately isolate the intrinsic noise of the amplifier.

-
- [1] H. Nyquist, *Physical Review* **32**, 110 (1928).
 - [2] M. Sampietro, L. Fasoli, and G. Ferrari, *Review of Scientific Instruments* **70**, 2520 (1999).
 - [3] P. Horowitz and W. Hill, *The Art of Electronics*, 3rd ed. (Cambridge University Press, 2015).
 - [4] J. Folk, “Lecture note 7,” PHYS 309 Laboratory Course, University of British Columbia (2025), available on UBC Canvas (Lec10.23.25.pdf).
 - [5] P. Welch, *IEEE Transactions on Audio and Electroacoustics* **15**, 70 (1967).